THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A man of weight 800 N exerts 40000 Pa when wearing shoes. The area of contact of both shoes is: 1

(a) 0.02 square metre **(b)** 0.2 square metre
(c) 2 square metres **(d)** 32 square metres

2. Why does air pressure fall as altitude increases? 1

(a) less air lies above **(b)** temperature rises
(c) water vapour falls **(d)** air is heavier

3. Assertion (A): A cyclone loses strength after moving inland. Reason (R): Cyclones draw energy from warm sea water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. An ice skater of weight 500 N skates on a blade of area 0.0005 square metre. The pressure on the ice is: 1

(a) 1000000 Pa **(b)** 250 Pa **(c)** 0.1 Pa **(d)** 1000 Pa

5. During a cyclone, the sea level near the coast rises due to: 1

(a) tides alone **(b)** storm surge **(c)** melting ice **(d)** earthquakes

6. Assertion (A): Wind blows from a region of high pressure to a region of low pressure. Reason (R): Air moves so as to reduce the pressure difference. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A pool is 2 m deep at one end and 4 m deep at the other. Compared to the shallow end, the pressure due to the water at the bottom of the deep end is: 1

(a) four times **(b)** twice **(c)** the same **(d)** half

8. Which instrument will warn of increasing wind speed in a storm? 1

(a) barometer **(b)** thermometer **(c)** anemometer **(d)** hygrometer

9. **Assertion (A):** Liquids exert pressure only on the base of the container. 1
Reason (R): Liquids exert pressure in all directions.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Air exerts pressure on everything in it. 1
Reason (R): Atmospheric pressure decreases as we go to greater heights.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do cyclones spin anticlockwise in the northern hemisphere? 2

12. Why do mountaineers need to acclimatise before climbing at high altitude? 2

13. Explain why hot air rises and what this has to do with winds. 2

14. Explain why a sealed packet of biscuits bulges when taken to a hill station. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a suction cup is held to a wall and why it falls if the wall is wet and cracked. 3

16. Why do deep-sea fish die when brought quickly to the surface? 3

17. Why does standing on loose snow with normal shoes make you sink while snowshoes help? 3

OR

The pressure at a point 3 m under water is P. Find the pressure at 9 m and justify. 3

18. Explain how the shape of a tornado funnel leads to extreme damage. 3

OR

Why does water not run out of an inverted glass covered by a card? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A cyclone formed over the Bay of Bengal and moved towards the Odisha coast. Meteorologists reported very low pressure at the centre, a calm region called the eye, and strong winds in the surrounding wall. Authorities moved people from the coast and a storm surge flooded low villages. After landfall the cyclone weakened within a day. 5
(i) Where do tropical cyclones form?
(ii) Name the calm centre of a cyclone.
(iii) Why do winds rush towards the centre?
(iv) Give two reasons why the cyclone weakened over land.

20. Describe an experiment to show that air exerts pressure and explain it. Add two more demonstrations and a daily-life application. 5

OR

A deep-sea submersible descends to 500 m. Explain how pressure changes, why walls must be strong and what safety checks are needed. 5

END OF QUESTION PAPER — **Total: 40 Marks**